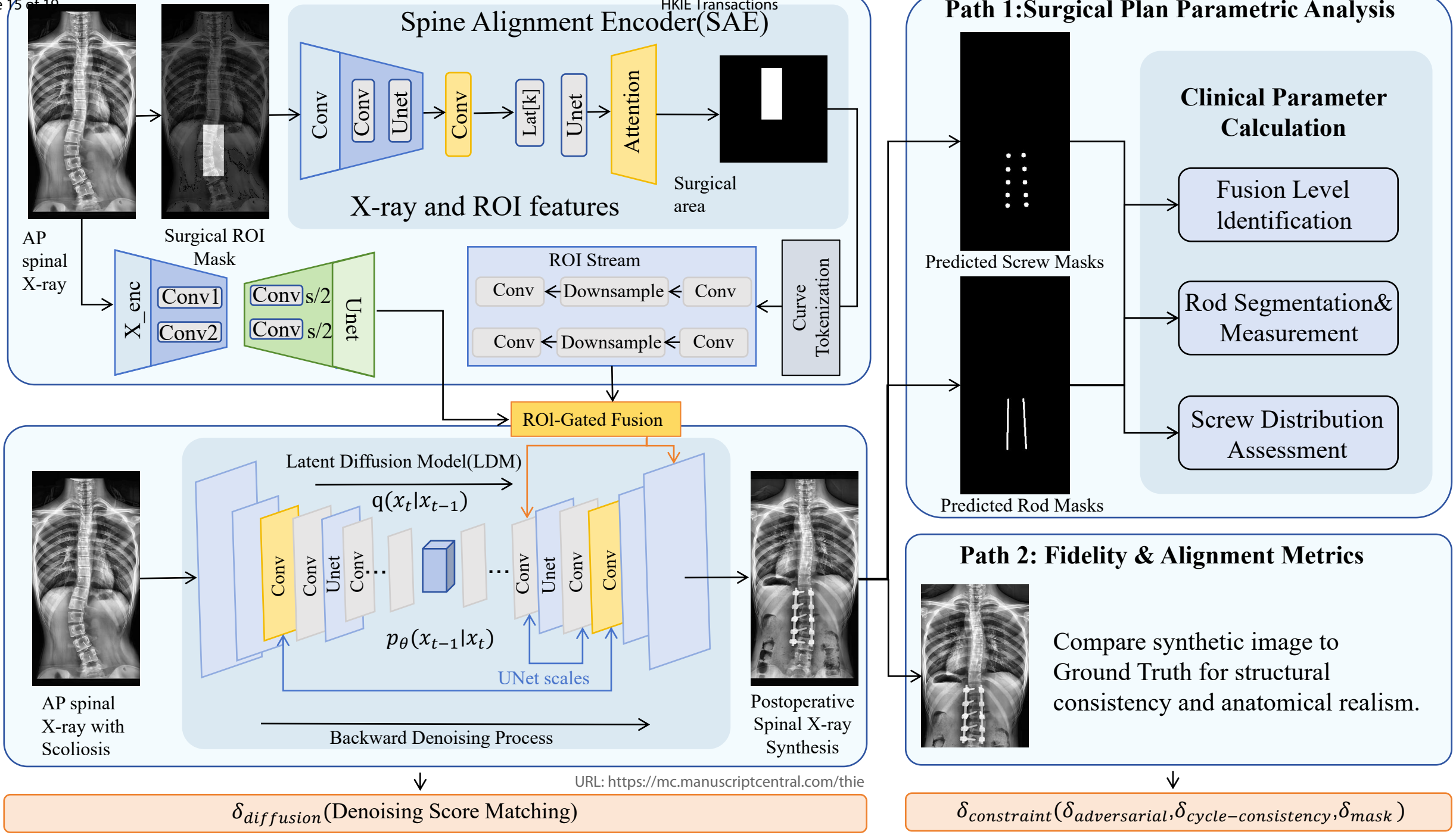

Spine Alignment Encoder(SAE)
AP spinal X-ray
Surgical ROI Mask
Conv
Conv
Unet
Conv
Lat[k]
Unet
Attention
Surgical area
X-ray and ROI features
X_enc
Conv1
Conv2
Conv s/2
Conv s/2
Unet
ROI Stream
Conv
Downsample
Conv
Conv
Downsample
Conv
Curve Tokenization
ROI-Gated Fusion
Latent Diffusion Model(LDM)
$q(x_t|x_{t-1})$
$p_\theta(x_{t-1}|x_t)$
UNet scales
Backward Denoising Process
AP spinal X-ray with Scoliosis
Postoperative Spinal X-ray Synthesis
Path 1:Surgical Plan Parametric Analysis
Predicted Screw Masks
Predicted Rod Masks
Clinical Parameter Calculation
Fusion Level Identification
Rod Segmentation& Measurement
Screw Distribution Assessment
Path 2: Fidelity & Alignment Metrics
Compare synthetic image to Ground Truth for structural consistency and anatomical realism.
$\delta_{diffusion}$(Denoising Score Matching)
$\delta_{constraint}(\delta_{adversarial}, \delta_{cycle-consistency}, \delta_{mask})$